# Supplementary Material

FastSET: Fast Sequential Retraining for Online Adaptation of Neural Receivers via Physics-Guided Decomposition

The experiments use the 8–64 split and the main manuscript's training settings unless a different split or training parameter is specified. The end-to-end (E2E) receiver is the baseline. Batch-size and adaptation-trajectory comparisons use the stopping-validation samples; the training-set-size comparison uses the separate PEBS evaluation samples. Section VII evaluates final models with fixed weights on new samples independent of training, stopping, and split selection. Adaptation samples are generated separately. Bit error rate (BER) is evaluated using 3,200 frames and 26,054,400 information bits per signal-to-noise ratio (SNR) point. Adaptation time denotes computation time for input processing and training; Section IV separately estimates BER-checking overhead. BER axes are logarithmic.

## I. Impact of Batch Size

Batch size determines the number of frames processed in each iteration. To examine the sample requirement for adaptation, batch sizes of 16, 32, and 64 are compared on A2B using the same pretrained weights, shuffled frame sequence, and learning rate of $10^{-3}$. Each run records five estimator iterations followed by one demapper iteration. Training continues after the BER target is reached to compare BER at subsequent iterations.

All three batch sizes first reach BER $< 2 \times 10^{-4}$ after one estimator iteration (Figure 1), with BERs of $1.48 \times 10^{-4}$, $1.51 \times 10^{-4}$, and $1.52 \times 10^{-4}$ for batch sizes 16, 32, and 64, respectively. At iteration 6, the corresponding BERs are $1.39 \times 10^{-4}$, $1.64 \times 10^{-4}$, and $1.17 \times 10^{-4}$. A batch size of 16 reaches the A2B target with half the samples used by the default batch size of 32. Batch size 32 is retained across the main experiments, with the same stopping iteration observed at half and twice the batch size.

## II. Impact of Receiver Architecture on Source-Domain Performance

The source-domain BER of pretrained FastSET and E2E is evaluated on TDL-A. FastSET uses five estimator pretraining epochs and ten demapper pretraining epochs; E2E uses ten pretraining epochs, following the main experimental setup.

Figure 2 compares the pretrained receivers before adaptation. At 12 dB, FastSET and E2E achieve BERs of $1.93 \times 10^{-4}$ and $1.99 \times 10^{-4}$, respectively. The adaptation comparison therefore starts from receivers with comparable source-domain BER.

## III. Impact of Joint Adaptation and Initialization

The Joint comparison examines the online update strategy and the effect of pretraining. Joint (SP) shares FastSET's architecture and pretrained model, with five estimator epochs and ten demapper epochs, so sequential and joint online updates are compared under matched initialization. Joint (JP) uses ten epochs of joint pretraining. Both Joint variants update both

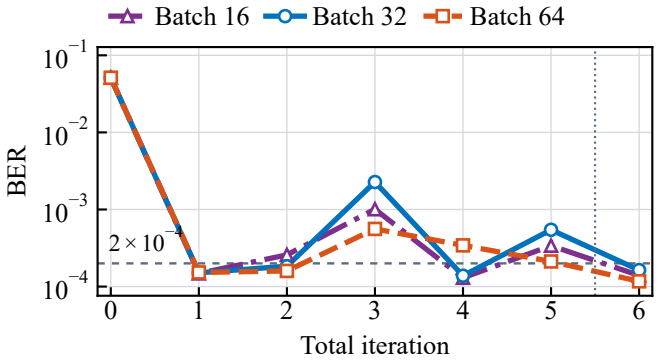



Fig. 1: BER with batch sizes 16, 32, and 64 on A2B at 12 dB. Iterations 1–5 update the estimator; iteration 6 updates the demapper. The vertical guide marks the stage transition.

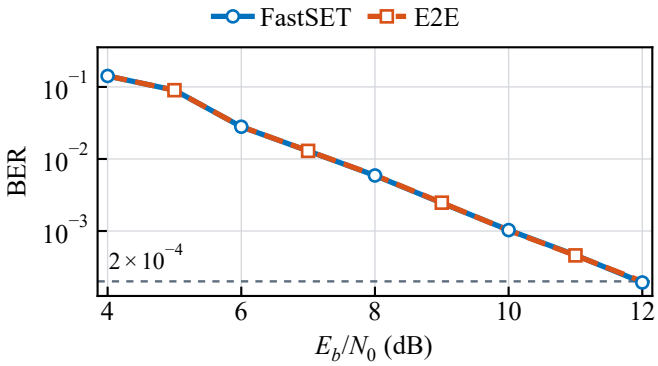



Fig. 2: Source-domain BER of pretrained FastSET and E2E on TDL-A before adaptation. The SNR range matches the main manuscript.

TABLE I: Mean adaptation time over three runs on A2B and A2C, recorded when BER first falls below $2 \times 10^{-4}$ at 12 dB.

| Method | A2B (s) | A2C (s) |
|---|---|---|
| FastSET (Est. 5) | 0.128 | 0.553 |
| Joint (SP) | 0.271 | 0.635 |
| Joint (JP) | 0.547 | 0.905 |
| E2E | 9.534 | 16.689 |

modules online by minimizing the sum of mean squared error (MSE) and binary cross entropy (BCE), $L_{\mathrm{MSE}} + L_{\mathrm{BCE}}$, with unit weights. MSE and BCE gradients update the estimator; BCE gradients update the demapper. Both variants use the same forward computation. Each module uses an Adam optimizer with states reset before adaptation.

All methods use a batch size of 32 and a learning rate of $10^{-3}$. Three runs use different online batch orders, with the same order used for all methods in each run. Training continues

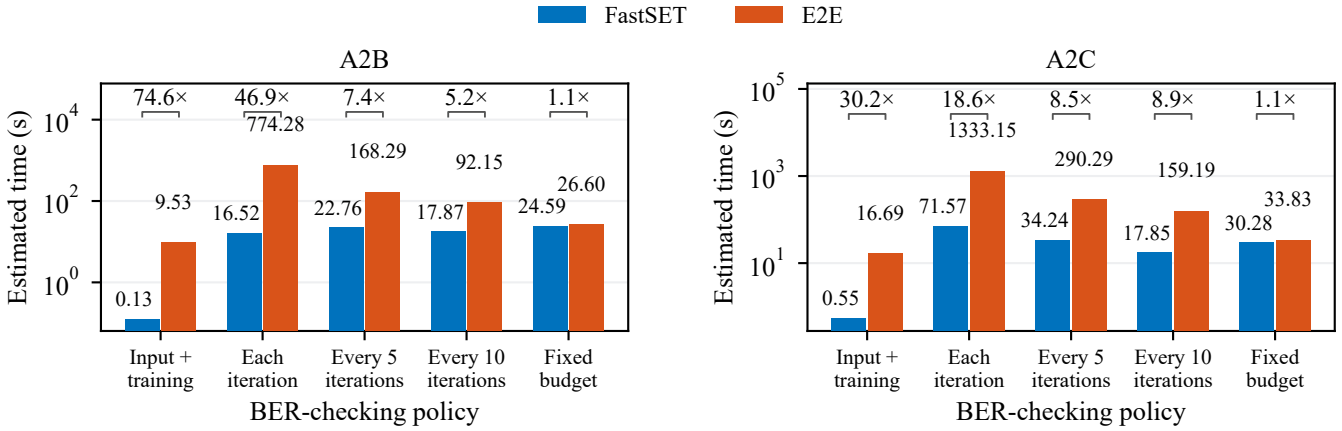



Fig. 3: Estimated adaptation time with four BER-checking schedules, compared with input processing and training alone. Logarithmic time axes show seconds; brackets give the E2E/FastSET time ratio. Each full BER check is assigned 16.39 s. Fixed budgets contain 50 iterations on A2B and 84 iterations on A2C.

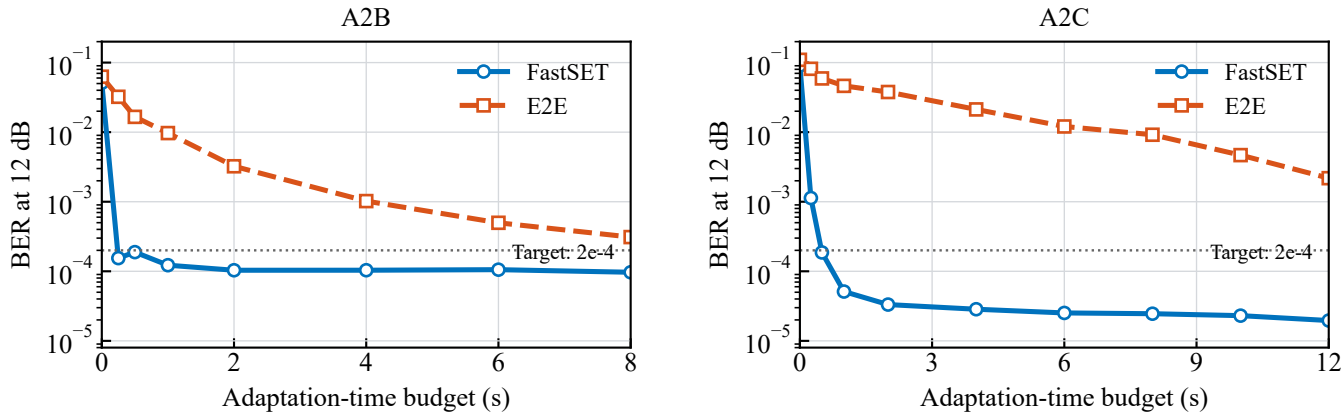



Fig. 4: BER at 12 dB under equal adaptation-time budgets. Each point averages three runs, using the last model saved within the budget in each run. FastSET performs five estimator iterations before demapper adaptation. Time includes input processing and training; the time axes are linear.

for 50 iterations on A2B and 84 iterations on A2C, including iterations after the BER target is reached. The receiver models, evaluation samples, and local timing procedure are retained from the main ablation experiments.

Joint (SP) reaches the A2B target after one iteration in every run, averaging 0.271 s; Joint (JP) requires two iterations and averages 0.547 s. On A2C, Joint (SP) reaches the target at iterations 1, 1, and 5, averaging 0.635 s. FastSET (Est. 5) averages 0.128 s on A2B and 0.553 s on A2C (Table I).

On A2B, FastSET and Joint (SP) both reach the target after one iteration. Updating only the estimator reduces the cost of the successful iteration from 0.271 to 0.128 s. On A2C, Joint (SP) is faster than FastSET in two runs; FastSET has the lower mean adaptation time over three runs. Joint (JP) requires more mean adaptation time than Joint (SP) on both A2B and A2C.

## IV. Impact of BER-Checking Frequency

BER-checking frequency affects both evaluation overhead and the iteration at which adaptation stops. Figure 3 compares checking after every iteration, every five iterations, every ten iterations, and once at the end of a fixed budget. The recorded training sequences contain 50 iterations on A2B and 84 iterations on A2C. For periodic checks, the first evaluated model with BER below $2 \times 10^{-4}$ is selected; the final model is also checked. The estimates begin at the start of adaptation, after the initial BER check.

For a selected model at iteration $n$, the estimated time is $\Delta t_T(n) + N_{\text{check}}\bar{T}_{\text{check}}$, where $N_{\text{check}}$ counts the scheduled BER evaluations. Input and training times come from local replay measurements. The common checking cost, $\bar{T}_{\text{check}} = 16.39$ s, is the mean of 50 recorded E2E evaluations, each processing 3,200 frames at 12 dB, and is applied to both receivers. Figure 3 averages the estimates over three runs.

Increasing the checking interval from one to ten iterations reduces the E2E estimates from 774.28 to 92.15 s on A2B and from 1333.15 to 159.19 s on A2C. E2E requires more training iterations, so the longer interval removes more BER evaluations for E2E. The E2E/FastSET time ratio consequently falls from 46.9 to 5.2 on A2B and from 18.6 to 8.9 on A2C. FastSET reaches the validation target within the first few iterations and continues training until BER meets the target at a scheduled check. On A2B, checking every five iterations requires a second

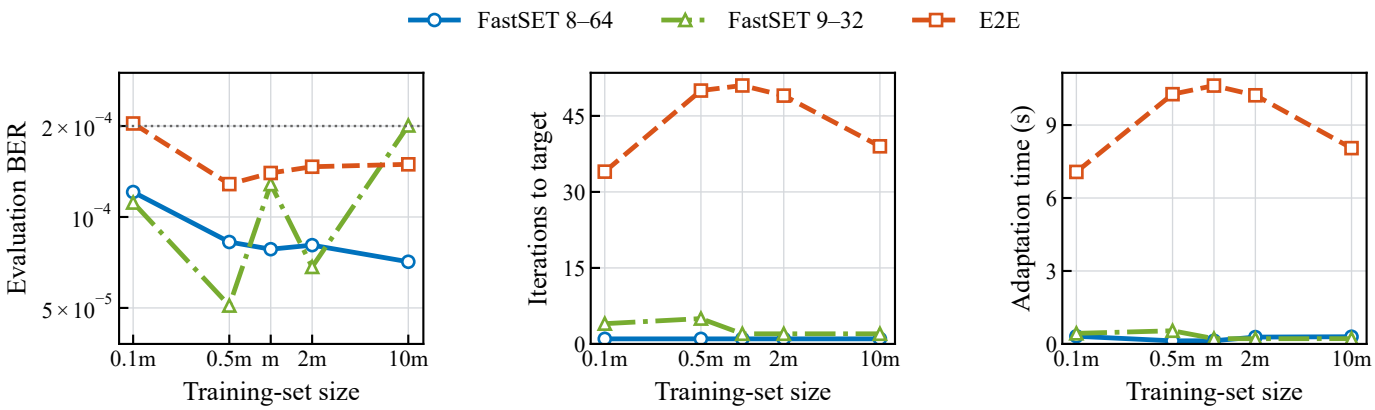



Fig. 5: Impact of online training-set size on A2B. Left: BER on the separate evaluation set. Middle: iteration count. Right: adaptation time. Results correspond to the first model with validation BER below $2 \times 10^{-4}$. The reference size is $m = 5,440$ frames. Training-set size and BER use logarithmic axes.

TABLE II: Independent evaluation of models saved when adaptation stops at 12 dB, using 3,200 frames (26,054,400 information bits) per receiver and target.

| Target | Receiver | Iterations | Bit errors | BER |
|---|---|---|---|---|
| A2B | FastSET | 1 | 7,858 | $3.02 \times 10^{-4}$ |
| A2B | E2E | 50 | 10,825 | $4.15 \times 10^{-4}$ |
| A2C | FastSET | 6 | 1,767 | $6.78 \times 10^{-5}$ |
| A2C | E2E | 84 | 10,629 | $4.08 \times 10^{-4}$ |

check in one run because the iteration-5 BER exceeds the target; checking every ten iterations uses one check in all three runs. Under fixed budgets, both receivers complete the entire recorded sequence and are checked once, giving time ratios of approximately 1.1. The fixed budgets retain many FastSET iterations after the target has first been reached. Checking more frequently allows adaptation to stop earlier, with additional evaluation overhead.

## V. Impact of Adaptation Time

The equal-time comparison examines the BER achieved when both receivers are allowed the same adaptation time. Figure 4 reports BER at 12 dB for models saved in the three-run ablation experiments. FastSET uses five estimator iterations followed by demapper iterations. Training continues beyond the BER target, covering 0–8 s on A2B and 0–12 s on A2C in every run.

Each point averages the BER of the last model saved within the budget in each run. Before the first iteration completes, the pretrained receiver is used. Time includes input processing and training and excludes BER evaluation.

With a 2 s budget, FastSET and E2E achieve BERs of $1.04 \times 10^{-4}$ and $3.23 \times 10^{-3}$ on A2B, and $3.32 \times 10^{-5}$ and $3.78 \times 10^{-2}$ on A2C, respectively. FastSET achieves lower BER at every plotted positive time budget. The BER advantage is therefore retained when both receivers continue training beyond the stopping target under equal time budgets.

## VI. Impact of Online Training-Set Size

The online training-set size determines how many frames are available for adaptation. The 8–64 split, the 9–32 split, and E2E are compared on A2B with sizes $0.1m$, $0.5m$, $m$, $2m$, and $10m$, where $m = 5,440$ frames. For each receiver, the source-pretrained weights, batch size of 32, and learning rate of $10^{-3}$ are held fixed. Figure 5 reports BER on the separate evaluation set, iteration count, and adaptation time for the first model that meets the validation BER target.

Across all five training-set sizes, the 8–64 split reaches the target after one estimator iteration using 32 frames, and the BER on the separate evaluation set remains below $2 \times 10^{-4}$. The 9–32 split requires two to five iterations, compared with 34–51 iterations for E2E. Both splits retain lower adaptation time than E2E across the evaluated sizes. The 8–64 split uses 32 frames before stopping from a pool containing at least 544 frames. Most available frames remain unused, explaining why increasing the training-set size has little effect on the adaptation time of the selected split.

## VII. Impact of Independent Test Samples

The final A2B and A2C models from the main performance comparison are evaluated on a new set, $\mathcal{D}_{\mathrm{ind}}$. The set contains 3,200 independently generated frames per target at 12 dB, using the same channel parameters as the main experiments. Both receivers process the same samples. The model weights are fixed before generating the set, which is used only for final evaluation. The recorded stopping iterations and receiver weights are retained.

Table II reports lower test BER for FastSET on both transitions. On A2B, both receivers exceed the validation target of $2 \times 10^{-4}$ on the independent samples. On A2C, FastSET remains below the target and E2E exceeds the target. The main adaptation times therefore describe computation up to the first validation threshold crossing. The independent results measure reception performance after the selected weights are fixed.